

CLIMB at VBS 2027: Multi-Signal Keyframe Retrieval with Collaborative Ad-Hoc Search

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Abstract. We present CLIMB, an interactive video retrieval system built for the Video Browser Showdown 2027. Because a single visual embedding often lacks the precision needed to capture cues, such as scene text, signage, or spoken dialogues, CLIMB retrieves information across four complementary signals. Visual embeddings, on-screen text, generated captions, and speech transcripts, which then are fused by rank not score and a solid mix between lexical and semantic matching depending on the signal used, ensure broad retrieval coverage.

Queries can be chained into temporal sequences, with each stage represented as an independent query, providing increased flexibility while reducing the search space.

For ad-hoc video search, CLIMB treats a task as a single shared team activity, hiding the shots, already submitted from everybody's results. Importantly search, browsing, and submission continue to work, even when the shared service is unreachable.

Keywords: Video Browser Showdown · Interactive Video Retrieval · Collaborative Search · Semantic Embedding.

1 Introduction

The Video Browser Showdown [9] is an interactive video retrieval competition asking participants to solve known-item search (KIS), ad-hoc video search (AVS), and visual question-answering (VQA/QAS) tasks against large scale video collections. For VBS 2027, the main collection is V3C [10] with 28,450 videos, roughly 3,800 hours, and about 4.7 TB, segmented into 4.14 million master shots, alongside the smaller MVK [12] and GynSurgLHE [6] collections.

Successfully addressing these tasks presents two fundamental challenges. First, a single visual embedding is often too vague to capture precise task hints, such as specific on-screen text or spoken dialogue. Second, because AVS scoring rewards distinct correct shots, a team of n operators searching independently spends a substantial share of its submissions re-finding what a teammate has already sent.

CLIMB addresses both. It indexes every video as scenes and keyframes, retrieves over four independent signals, fuses them by rank, and treats an AVS task as a single shared activity rather than n parallel ones, vanishing duplicate scenes from everybody's list. Concretely, we contribute:

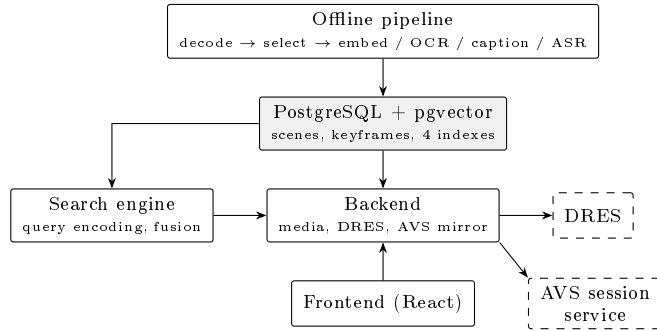


Fig. 1. CLIMB architecture. All components can run locally on each machine, except for two external services: DRES and the AVS session service, which is deployed once.

1. Keyframe-level retrieval fusing four heterogeneous signals by rank, with the choice between lexical and semantic matching made per signal (Sect. 4.1).
2. A temporal query language, `A >> B >>(120) C`, in which every stage is itself a full query (Sect. 4.3).
3. A collaborative, network-resilient AVS mode (Sect. 6).
4. An index engineered for everyday hardware: binary-quantized HNSW with oversample-and-rerank, fed by a resumable ingest pipeline (Sect. 3).

2 System Overview

CLIMB is built from six components (Fig. 1). Decoding each video, segmenting it into scenes, selecting keyframes, computing embeddings, reading text on screen, as well as captioning and transcribing, happens in an offline pipeline described in Sect. 3. The resulting scenes, keyframes, and four indexes that get queried by the retrievers land in a PostgreSQL database.

At query time the retrieval stack and main application run as separate processes, since they are written in different languages. While the models and fusion require a Python environment, media serving, DRES communication, and the AVS mirror are based on Node.js and Express. Both processes connect to the database: the backend reads the scene and keyframe metadata directly and only delegates embedding, ranking, and fusion to the search engine. The React frontend (Sect. 5), however, only talks to the backend.

For scenarios where network reliability is a concern, CLIMB supports fully localized deployments, serving media assets directly from localhost. Additionally the shared and optional AVS session service (Sect. 6) only transmits a few kilobytes of JSON polled every four seconds.

3 Indexing Pipeline

3.1 Scenes and Keyframe Selection

The V3C dataset provides scene boundaries. These scene boundaries are the center of our encoding. They get used, splitting the video into different scenes. For each of these scenes, we select between 2 and 32 keyframes according to the length of the scene, selecting one keyframe per four seconds.

We start by selecting the most typical frame and then repeatedly add the frame, which is most different from the selected keyframes so far. This selects a broad coverage of keyframes in a scene. Afterward we reject blank frames. A blank frame is one that is nearly flat and has almost no edges. This reduced the number of keyframes sampled from our subset (Sect. 7) from 99.661 keyframes at 1 fps to 35,558 keyframes. Damaged bitstreams are indexed from what we were able to decode, flagged, and logged.

3.2 Feature Extraction

CLIMB extracts four signals per scene, ensuring a divers set of retrieval possibilities, but whilst we have four different signals, they are deliberately not applied equally. OCR reads every keyframe, but captioning is only done once per scene. The model used is PP-OCR [2] via PaddleOCR where available and rapidocr-onnxruntime otherwise. For embedding a visual representation into a 1024-dimensional vector space, we use SigLIP2 [13].

Given the high computational cost and intra-scene visual redundancy, captioning is restricted to each scene’s first keyframe. This results in just 14,345 captions in comparison to 35,558 keyframes. We intend to leverage Qwen2-VL-2B-Instruct [15] for full-scale captioning, but used the lightweight SmolVLM-256M-Instruct [5] for testing. Audio transcription is handled by Whisper [7].

In order to encode textual signals into semantic embeddings, we use multilingual-e-base [14], embedding into a 768-dimensional vector space. For efficient retrieval, all generated embeddings are stored using an HNSW index [4] over binary-quantized vectors. To maintain search accuracy while minimizing memory overhead, initial search candidates are reranked against their full-precision vectors. This quantization significantly reduces the memory footprint, requiring only 5.3 GB for 12.4M keyframes compared to 28 GB for half-precision storage.

4 Retrieval Model

4.1 Rank-Based Fusion at Keyframe Level

As discussed before, we use four different encoders to encode as broad a spectrum of information as possible, but they signals do not produce comparable scores on their own. A small 20-query test on our subset returns a score of 0.17 for the best visual result (due to the modality gap described in [3]), whilst the best caption

match was a 0.93. In order to avoid sensitive and model specific calibration we use a ranked based fusion approach, explained below.

Each signal returns a list of ranked objects, scenes, or keyframes, respectively. Whilst the visual signal and OCR signal natively return a list of ranked keyframes, the caption and audio signals are scene-based. Thus, each keyframe inherits the rank of its parent scene.

The keyframes are then merged via weighted Reciprocal Rank Fusion (RRF) [1]. This works similarly to normal RRF but weighs the signals depending on their source, allowing us to highlight keyframes found via specific encoders, i.e., OCR, where a lexical hit is much more likely to belong to the correct scene than a similar visual embedding. The updated scoring formula is:

$$RRFscore(d \in D) = \sum_{r \in R} \frac{w(r)}{k + r(d)} \quad (1)$$

where D is the set of keyframes, R is the set of encoders, $w(r)$ is the weight of a specific encoder signal, k is a constant set to 60, and $r(d)$ is the rank the encoder r gave the keyframe d [1]. The constant k was chosen by the authors of the original RRF paper and helps to reduce the influence of outliers.

4.2 Semantic and Lexical matching

While cross-modal visual retrieval with SigLIP2 is inherently semantic, the remaining text-to-text signals could theoretically rely on lexical matching. However, strict keyword matching on generated captions suffers from severe vocabulary mismatch. For instance, a query for "law enforcement" shares zero lexical overlap with a highly relevant caption describing "a police uniform". In fact, the terms "law" and "enforcement" do not appear once within our extracted captions. Consequently, a purely lexical search would return an empty result set rather than just an inaccurate one.

While ASR transcripts are embedded semantically for conceptual and cross-lingual search, users can invoke a "said:" operator for exact lexical matching of specific quotes. Conversely, OCR operates as a purely lexical signal to pinpoint exact on-screen text provided in task hints. Semantic matching here is redundant, as general concepts are already captured by the visual embedding.

4.3 Query Language and Temporal Chains

Some retrieval tasks require describing not a single scene, but a temporally ordered sequence of scenes, such as a bridge followed by a beach. A plain text query searches all four signals at once, and the operator prefixes in Table 1 restrict it to a single source, whilst `>>` joins queries into ordered temporal chains.

The chaining makes the prefixes much more powerful. A query `A >> B >> C` is 3 stages, where each one is a complete query on its own. `text: pegasus >> mountain bike race` finds a scene with a text that reads "pegasus", followed by a scene that matches with mountain bike race, independent of a single encoder.

Table 1. Query syntax. Prefixes restrict searches to specific signals, while >> chains queries temporally within a single video.

Input	Effect
<code>a drawing of people on a beach</code>	Global search (all signals)
<code>text: state capitol</code>	On-screen text only
<code>said: new york</code>	Spoken audio only
<code>-video:00191</code>	Exclude a video (repeatable)
<code>a video game sunset >> door</code>	Sequential search (same video)

An unannotated chain searches for scenes in the same video that are at maximum 60 seconds apart, annotated with `(120)` or `(d120)`; it’s 120 seconds. The maximum delta is ten minutes, with up to four stages per chain.

5 Interactive Search

5.1 The Search Loop

A search typically follows an iterative interaction loop. The operator enters a query, which returns a grid of keyframes. Selecting one, displays a five-second scene preview starting at the corresponding timestamp. The filmstrip located under the video player, provides an overview of all the keyframes allowing for efficient navigation between them. Keyframe times get populated automatically but remain editable, to fine-tune the target submission frame. Additionally, there is a variety of button based controls for quick search refinement. To keep this cycle responsive, query results are cached and paginated: the initial query executes the search, while subsequent pages are lazily loaded from cache upon scrolling.

For task submission and search refinement (Fig. 2), *Submit to DRES* sends the selected timestamp. For false positives, *Exclude video from search* removes that video from the results. When textual queries fall short, *Find Similar* enables visual query-by-example. Finally, dedicated UI inputs support VQA tasks, allowing operators to submit text-only or combined text-and-shot answers.

5.2 Submitting to DRES

As with every browser at VBS, the DRES-Submission pipeline is an integral part of CLIMB [8, 11]. A login dialog can be opened, where the *server URL*, *evaluation name*, and *credentials* can be entered. If the requested evaluation is missing, CLIMB falls back to the first available and alerts the operator.

Every submission returns a status, consisting of a status code, summary, and verdict. This gets displayed to the user after the *Submit to DRES* button. Correct submissions are indicated in green, pending ones in yellow, and incorrect or rejected ones in red.

AVS submissions follow a different processing path, as described in Sect. 6. From the operator’s perspective, however, the interaction is analogous to that

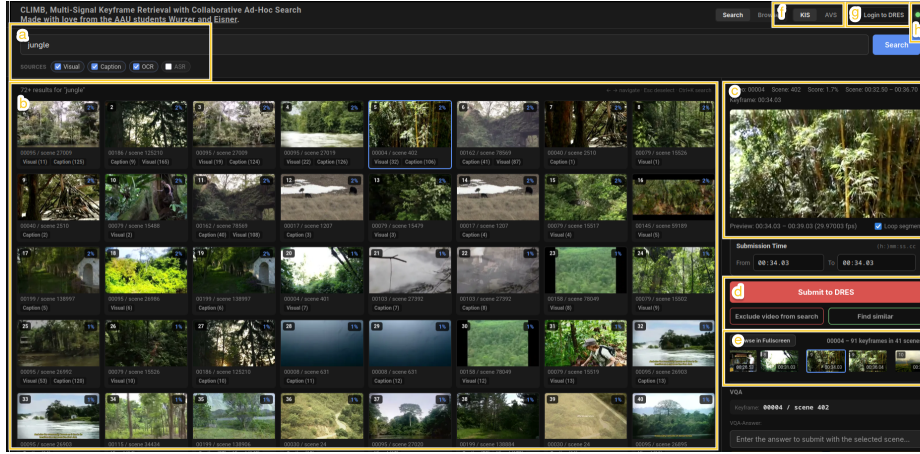


Fig. 2. The CLIMB interface: (a) query bar and source filters, (b) scene result grid, (c) detail panel and video player, (d) submission and search-refinement controls, (e) keyframe filmstrip, (f) AVS session tab, (g) DRES connection controls, and (h) system health status.

of a default KIS task. After switching to the AVS tab in the top-right corner of the interface and joining a session, submissions can be made through the same interaction pattern. The primary difference is that accepted submissions are hidden from other participants in the same session.

6 Collaborative Ad-Hoc Video Search

Since AVS scoring rewards distinct correct shots, independently searching team members tend to identify the same obvious results first, resulting in a large fraction of duplicate submissions. We address this problem by making an AVS session a shared state. Submitted scenes are consequently hidden from the results of all session participants. Other scenes in these videos get marked as covered, but not hidden to promote a diverse submission set, whilst allowing for submission of multiple distinct scenes from the same video.

Importantly, search and submissions do not depend on the speed or availability of the AVS collaboration service. To ensure this separation, CLIMB sends the submission request to the DRES server before attempting to communicate with the, independently deployed, collaboration service.

7 Preliminary Results

The pipeline is implemented for the full V3C collection; the numbers below were measured on a 200-video subset available at the time of writing (25 h, 14,345 shots, 35,558 keyframes).

Compared with exhaustive exact search over 7,543 keyframes (108 ms), over-sampling 1,000 candidates before exact reranking recovers 15 of the true top 20 in 29 ms, 2,000 candidates recover 18 in 54 ms, and 4,000 all 20 in 91 ms. The remaining discrepancy is due to the binary quantization rather than the index: the ANN stage ranks by Hamming distance over 1-bit vectors. CLIMB defaults to 1,000 candidates, while keeping this parameter configurable.

A far simpler version of CLIMB won our university’s course-internal retrieval competition preceding this submission. However we expect VBS to be considerably more challenging. Query latency has been measured at 7,543 keyframes rather than the about 4M shots of the full collection, and neither MVK nor GynSurg resembles the imagery SigLIP2 was trained on.

8 Conclusion

We presented CLIMB, an interactive video retrieval system for VBS 2027. It fuses four heterogeneous signals, namely visual embeddings, on-screen text, generated captions, and transcripts, at the keyframe level using rank-based fusion. Operators can further chain multiple queries into temporally ordered sequences.

Its collaborative mode is designed around the AVS scoring rule. Shots a teammate has already submitted disappear from everyone’s results, while search, browsing, and submission remain unaffected when the shared session service is unavailable.

Our next steps are to complete ingestion of the full collection and address the domain gap for MVK and GynSurg, where the general-purpose visual embeddings provided by SigLIP2 are expected to be least effective.

References

1. Cormack, G.V., Clarke, C.L.A., Buettcher, S.: Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In: *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*. p. 758–759. SIGIR '09, Association for Computing Machinery, New York, NY, USA (2009). <https://doi.org/10.1145/1571941.1572114>, <https://doi.org/10.1145/1571941.1572114>
2. Li, C., Liu, W., Guo, R., Yin, X., Jiang, K., Du, Y., Du, Y., Zhu, L., Lai, B., Hu, X., et al.: Pp-ocrv3: More attempts for the improvement of ultra lightweight ocr system. *arXiv preprint arXiv:2206.03001* (2022)
3. Liang, V.W., Zhang, Y., Kwon, Y., Yeung, S., Zou, J.: Mind the gap: Understanding the modality gap in multi-modal contrastive representation learning. In: Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., Oh, A. (eds.) *Advances in Neural Information Processing Systems*. vol. 35, pp. 17612–17625. Curran Associates, Inc. (2022). <https://doi.org/10.52202/068431-1280>, https://proceedings.neurips.cc/paper_files/paper/2022/file/702f4db7543a7432431df588d57bc7c9-Paper-Conference.pdf

4. Malkov, Y.A., Yashunin, D.A.: Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **42**(4), 824–836 (2020). <https://doi.org/10.1109/TPAMI.2018.2889473>
5. Marafioti, A., Zohar, O., Farré, M., Noyan, M., Bakouch, E., Cuenca, P., Zakka, C., Allal, L.B., Lozhkov, A., Tazi, N., et al.: Smolvlm: Redefining small and efficient multimodal models. *arXiv preprint arXiv:2504.05299* (2025)
6. Nasirihaghighi, S., Ghamsarian, N., Peschek, L., Munari, M., Husslein, H., Sznitman, R., Schoeffmann, K.: Gynsurg: A comprehensive gynecology laparoscopic surgery dataset. In: *Proceedings of the 33rd ACM International Conference on Multimedia*. p. 13141–13147. MM '25, Association for Computing Machinery, New York, NY, USA (2025). <https://doi.org/10.1145/3746027.3758267>, <https://doi.org/10.1145/3746027.3758267>
7. Radford, A., Kim, J.W., Xu, T., Brockman, G., Mcleavey, C., Sutskever, I.: Robust speech recognition via large-scale weak supervision. In: Krause, A., Brunskill, E., Cho, K., Engelhardt, B., Sabato, S., Scarlett, J. (eds.) *Proceedings of the 40th International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 202, pp. 28492–28518. PMLR (23–29 Jul 2023), <https://proceedings.mlr.press/v202/radford23a.html>
8. Rossetto, L., Gasser, R., Sauter, L., Bernstein, A., Schuldt, H.: A system for interactive multimedia retrieval evaluations. In: Lokoč, J., Skopal, T., Schoeffmann, K., Mezaris, V., Li, X., Vrochidis, S., Patras, I. (eds.) *MultiMedia Modeling*. pp. 385–390. Springer International Publishing, Cham (2021)
9. Rossetto, L., Schoeffmann, K., Gurrin, C., Lokoc, J., Bailer, W.: Results of the 2025 video browser showdown. *ArXiv abs/2509.12000* (2025), <https://doi.org/10.48550/arXiv.2509.12000>
10. Rossetto, L., Schuldt, H., Awad, G., Butt, A.A.: V3c – a research video collection. In: Kompatsiaris, I., Huet, B., Mezaris, V., Gurrin, C., Cheng, W.H., Vrochidis, S. (eds.) *MultiMedia Modeling*. pp. 349–360. Springer International Publishing, Cham (2019)
11. Sauter, L., Gasser, R., Schuldt, H., Bernstein, A., Rossetto, L.: Performance evaluation in multimedia retrieval. *ACM Transactions on Multimedia Computing, Communications and Applications* (Oct 2024). <https://doi.org/10.1145/3678881>, <https://doi.org/10.1145/3678881>
12. Truong, Q.T., Vu, T.A., Ha, T.S., Lokoč, J., Wong, Y.H., Joneja, A., Yeung, S.K.: Marine video kit: A new marine video dataset for content-based analysis and retrieval. In: Dang-Nguyen, D.T., Gurrin, C., Larson, M., Smeaton, A.F., Rudinac, S., Dao, M.S., Trattner, C., Chen, P. (eds.) *MultiMedia Modeling*. pp. 539–550. Springer International Publishing, Cham (2023)
13. Tschannen, M., Gritsenko, A., Wang, X., Naeem, M.F., Alabdulmohsin, I.M., Parthasarathy, N., Evans, T., Beyer, L., Xia, Y., Mustafa, B., H'enaiff, O., Harmsen, J., Steiner, A., Zhai, X.Q.: Siglip 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features. *ArXiv abs/2502.14786* (2025), <https://arxiv.org/abs/2502.14786>
14. Wang, L., Yang, N., Huang, X., Yang, L., Majumder, R., Wei, F.: Multilingual e5 text embeddings: A technical report. *arXiv preprint arXiv:2402.05672* (2024)
15. Wang, P., Bai, S., Tan, S., Wang, S., Fan, Z., Bai, J., Chen, K., Liu, X., Wang, J., Ge, W., et al.: Qwen2-vl: Enhancing vision-language model's perception of the world at any resolution. *arXiv preprint arXiv:2409.12191* (2024)